

ANDERSON JUNIOR COLLEGE
2008 JC2 PRELIMINARY EXAMINATION

MATHEMATICS
Higher 1

8863/01

Paper 1

9 September 2008

0800 hr – 1100 hr

3 hours

Additional Materials : Answer Paper
 Graph paper
 List of Formulae (MF15)

READ THESE INSTRUCTIONS FIRST

Write your name and class on the cover page and all the work you hand in..

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer all the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use a graphic calculator.

Unsupported answers from a graphic calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question.

At the end of the examination, fasten all your work securely together with this cover page on top.

Question No.	Marks
1	/ 3
2	/ 2
3	/ 7
4	/ 8
5	/ 9
6	/ 11
7	/ 7
8	/ 6
9	/ 7
10	/ 8
11	/ 12
12	/ 9
13	/ 11
Total	/ 100

Name: _____

PDG: ____/07

Section A: Pure Mathematics [40 marks]

- 1 Solve the simultaneous equations

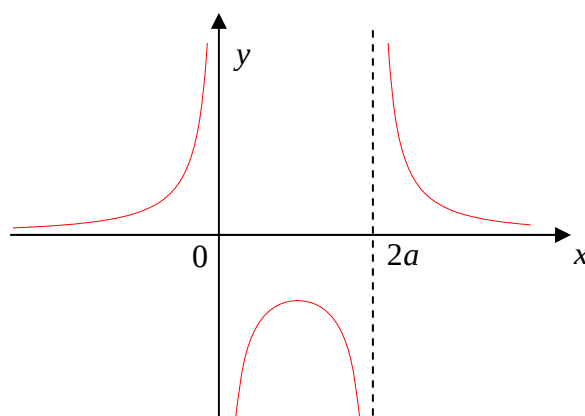
$$\log_2 xy = 7$$

$$\log_2 \frac{x^2}{y} = 5.$$

[3]

- 2 The diagram shows the graph of $y = f(x)$ with maximum turning point $(a, -b)$ where a and b are positive real numbers. Sketch the graph of $y = f'(x)$.

[2]



- 3 (a) Differentiate $[\ln(2x)]^5$ with respect to x

[2]

- (b) (i) Find $\int 4 - e^{2x} dx$

[1]

- (ii) The equation of the tangent to the curve of $y = 4 - e^{2x}$ at the point where $x = 0$ is given by $y + 2x = 3$. Write down the exact coordinates of the point where the curve crosses the x -axis and hence show that the area bounded by the curve, tangent and x -axis is given by $\frac{15}{4} - 4 \ln 2$.

[4]

- 4 (a) Solve the inequality $|x| < 4|x - 3|$.

[2]

- (b) The quadratic equation $(k + 1)x^2 + 12x + (k - 4) = 0$ has real roots.

- (i) Show that $k^2 - 3k - 40 \leq 0$.

[2]

- (ii) Solve the inequality in (i).

[2]

- (iii) Hence solve the inequality $\tan^2 \beta - 3 \tan \beta - 40 \geq 0$ for $-\pi \leq \beta \leq \pi$.

[2]

5 The functions f and g are defined by

$$f : x \rightarrow x^2, \quad x \in \mathbb{R},$$
$$g : x \rightarrow \frac{1}{x+2}, \quad x \in \mathbb{R}, x \neq -2,$$

- (i) State the range of g . [1]
- (ii) Write down $fg(x)$. [1]
- (iii) Sketch the graph of $y = fg(x)$, showing clearly any asymptotes. State the exact coordinates of any points where the graph crosses the axes. [2]
- (iv) Explain why the function fg does not have an inverse. [1]
- (v) Suggest a maximal restricted domain of fg such that the function fg has an inverse. [1]
- (vi) Based on the domain you have chosen in (iv), find $(fg)^{-1}(x)$. [3]

6 (a) At the point (x, y) , where $x > 0$, the gradient of a curve is given by

$$\frac{dy}{dx} = 3x^{\frac{1}{2}} + \frac{16}{x^2} - 7$$

- (i) Verify that $\frac{dy}{dx} = 0$ when $x = 4$ and explain briefly how you can tell that the point where $x = 4$ is a minimum or maximum. [2]
 - (ii) The point $P(1, 8)$ lies on the curve.
Find an equation of the normal to the curve at P . [2]
 - (iii) Find the equation of the curve which passes through the point $P(1, 8)$. [3]
- (b) The length of the side of a square, x cm, is increasing at a constant rate of 1.2 cm s^{-1} . At the instant when the length of the side is 10 cm, find the rate of increase of the
- (i) perimeter, [2]
 - (ii) area [2]

Section B: Statistics [60 marks]

- 7 In a certain country, it is known that 80% of the adult population are right-handed, 15% are left-handed and 5% are ambidextrous (i.e. able to use either hand equally well).

If 10 adults from this country are selected at random, find the probability that

- (i) at least one of those chosen will be ambidextrous [2]
- (ii) less than three of those chosen will be left-handed. [2]

Using a suitable approximation, find the probability that in a randomly chosen sample of 100 adults, 85 or more will be right-handed. [3]

- 8 A bakery produces large and small loaves of bread with masses, in g, that are normally distributed with means and standard deviations as shown in the following table.

Size of bread	Mean mass	Standard deviation
Large	420	30
Small	220	10

- a) Find the probability that one large loaf weighs more than twice the weight of a small loaf. [3]
- b) Find the probability that the total weight of 5 large loaves and 10 small loaves lies between 4.25 kg and 4.4 kg. [3]

- 9 A university is hosting a summer school for 2000 students. All the students are accommodated in Halls of Residence on the campus. An administrator has been asked to conduct a survey to measure satisfaction with the quality of life on campus. A sample of students is to be obtained.

- (a) The administrator has an alphabetical listing of the names of all 2000 students and they are numbered consecutively, from 0000 to 1999. A number between 0 and 24, inclusive, is selected at random. The corresponding student and every 25th student thereafter is included in the sample.

- (i) What is the name given to this method of sampling? [1]
 - (ii) Write down the size of the sample that will be obtained. [1]
 - (iii) Give an advantage of using this method of sampling rather than simple random sampling. [1]

- (b) The following table contains a breakdown of the students by gender and year of study.

Gender	Year 1	Year 2	Year 3	Year 4	Total
Female	360	260	240	240	1100
Male	280	240	200	180	900
Total	640	500	440	420	2000

A stratified random sample of size 100 is required.

- (i) Construct a table as above to indicate the appropriate number of males and females from each year of study to be included in the sample. [3]
- (ii) Give one advantage that stratified sampling has over simple random sampling. [1]

- 10 Mary belongs to an athletic club. In javelin throwing competitions 85% of her throws exceed 35 m and 70% exceed 37.5 m. Her throws are normally distributed.
- (i) Find the mean and standard deviation of Mary's throws, each correct to two significant figures.
[3]

- (ii) What distance will be exceeded by 60% of her throws? [1]

Jane belongs to the same club and in competitions, her throws are normally distributed with mean 41.0 m and standard deviation 2.0 m.

- (iii) What is the probability that, in a competition in which each athlete takes single throw, Jane will beat Mary?
[2]

The club has to choose one of these two athletes to enter a major competition. In order to qualify for the final rounds it is necessary to achieve a throw of at least 49 m in the preliminary rounds. Which one should be chosen and why? [2]

- 11 A cluster of flats consists of 320 apartments: 200 are four-room type and 120 five-room type. The number of children living in these apartments are shown in the table

	Number of children			
	None	One	Two	At least three
4-Room	40	37	88	35
5-Room	24	32	41	23

An apartment in the cluster is selected at random.
 Q denotes the event "the apartment is a four-room type"
 R denotes the event "no children live in the apartment"
 S denotes the event "one child lives in the apartment"
 T denotes the event "two children live in the apartment"

- (a) Find
- (i) $P(Q \cap R)$ [1]
- (ii) $P(Q \cup T)$ [1]
- (iii) $P(Q|R)$ [2]
- (iv) $P(R|Q')$ [2]
- (b) (i) Name two of the events Q, R, S and T that are mutually exclusive. [1]
- (ii) Determine whether the events Q and R are independent. Justify your answer. [2]
- (c) Define, in the context of this question, the event
- (i) $Q' \cup T$. [1]
- (ii) $Q \cap (R \cup S)$ [2]

- 12 The handicap committee of a golf club has indicated that the mean score achieved by the club's members in the past was 85.9.
A group of members believes that recent changes to the golf course have led to a change in the mean score achieved by the club's members and decides to investigate this belief.
A random sample of the scores, x , of 100 club members was taken and is summarized by

$$\sum (x - 85) = -150 \text{ and } \sum (x - 85)^2 = 15546$$

where $\bar{x}$ denotes the sample mean.

- (i) Find unbiased estimates of the population mean and variance. [3]
 - (ii) Assuming a normal distribution, test at the 5% significance level the group's belief that the mean score has changed. [4]
 - (iii) Another group of members believes that the mean score has decreased as a result of the changes. If the sample data remains unchanged, what is the largest level of significance that will lead to the rejection of the belief of this group, giving your answer to the nearest integer? [2]
- 13 Cucumbers are stored in brine before being processed into pickles. Data were collected on x , the percentage of sodium chloride in the salt used to make brine, and y , a measure of the firmness of the pickles produced. The data are shown below:

x	6.0	6.5	7.0	7.5	8.0	8.5	9.0	9.5
y	15.3	15.8	16.1	16.7	17.4	17.8	18.2	18.3

- (a) Give a sketch of the scatter diagram for the data, as shown on your calculator. [1]
- (b) Calculate the product moment correlation coefficient and comment on the value obtained. [2]
- (c) Calculate the equation of the regression line of y on x . [1]
- (d) Interpret in context the value of the gradient of the regression line. [1]
- (e) Comment on the value of the intercept with the y -axis of the regression line. [1]
- (f) Explain why in this case it would be inappropriate to calculate the equation of the regression line of x on y . [1]
- (g) Use the equation of the line to predict the value of y when x is
 - (i) 6.7 [1]
 - (ii) 10.7 [1]
 Comment briefly on these predictions. [2]

